Ayush Urs

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Education

University of California, San Diego — San Diego, CA
MS Computer Science — *September 2026 – 2028*

University of California, San Diego — San Diego, CA
B.S. Mathematics – Computer Science — *GPA: 3.89* *September 2023 – March 2026*
Relevant Coursework: Machine Learning: Algorithms, Data Structures, Stochastic Processes, Recommender Systems

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, Bash, Dart, MATLAB, Stata
ML & AI: PyTorch, TensorFlow, LangChain, Keras, scikit-learn, OpenCV; LLM fine-tuning (LoRA, DPO), RAG, vector search, LLM agents, MCP (Model Context Protocol), SVM, NLP, CNN, RNN, Hugging Face
Systems & Data: Kubernetes, MongoDB, CI/CD, Vertex AI, AWS (EC2, ECS/Fargate), Docker, PostgreSQL, SQLite, pandas, NumPy, Apache Spark, Databricks, multi-node GPU training, Git, Jupyter, VS Code, Claude code, FastAPI

Experience

AI Researcher — Remote
University of Wisconsin–Madison — *June 2026 – Present*
- Engineered a **7-agent LangChain** pipeline with **Abstract Syntax Tree** validation that converts **20+** research papers into runnable biological simulations, reducing experiment time from **15 to 6 minutes**.
- Standardized **25+** simulation tools through an **MCP server**, and added **30+ CI/CD regression tests** to catch breaking changes before execution cutting integration overhead **40%**.

Machine Learning Researcher — San Diego, CA
Halıcıoğlu Data Science Institute — *March 2026 – June 2026*
- Designed a **bandit-based selection strategy** with **divide-and-conquer decomposition** for **LLM-driven algorithm discovery**, benchmarked across **Qwen**, **Mistral**, and **Llama** models, building on the open-source **SkyDiscover** framework and outperforming EoH and FunSearch on compute efficiency.
- Evaluated the strategy across **200+** optimization tasks on a SQLite-backed strategy database: **31%** fewer LLM calls, **24%** faster convergence.

Machine Learning Researcher — San Diego, CA
Human-centered eXtended Intelligence, UCSD Lab — *September 2025 – Present*
- Built Reddit-based pipelines to generate **65K+ preference pairs** on **Vertex AI (GCP)** for **CERQ-aligned** emotion regulation, improving classification by **21** percentage points.
- Fine-tuned transformer models using **DPO** on **4x A100 Nautilus clusters** across **100K+** samples, with responses preferred in **72%** of human evaluations for regulated vs dysregulated variants. **Second Author**

Machine Learning Intern — New York, NY (Hybrid)
Myna Voice Labs (Research Collaboration with Princeton PhD Scholars) — *August 2025 – April 2026*
- Achieved **89%** accuracy with cross-linguistic generalization, developing and deploying Parkinson's disease detection models (**SVMs, CNNs, Audio Spectrogram Transformers**) via **GCP Vertex AI** pipelines.
- Evaluated model robustness against audio perturbations using **Bayesian confidence intervals** across **OpenL3** and **Whisper** feature representations, finding a **20–28x** separation in interval width between class predictions; added **LoRA-based continual learning** that retained **95.6%** of prior-task performance.

Publications & Projects

Can We Break LLMs Out of Self-Loops? Fine-Grained Reasoning Control with Activation Steering. Submitted, **EACL**.

Building an AI-Enhanced Organic Island with Brassica Juncea Seeds to Counter Eutrophication. Published, **JSR**.

Skin Lesion Detection System | *PyTorch, YOLOv8, Docker, Kubernetes*
- Contributed to an end-to-end skin lesion classification pipeline on **HAM10000**, benchmarking a custom **CNN** against **EfficientNet-B0**, **ResNet18**, and **YOLOv8s**, reaching **81.75%** top-1 accuracy.
- Built **GPU training infrastructure** with **Docker** and **Kubernetes** on **A100** nodes.

RAG Pipeline for Internal Query Answering | *LangChain, sentence-transformers, MongoDB, FastAPI*
- Built **RAG** pipeline to replace manual internal doc search, using **LangChain** and **sentence-transformers**.
- Built the data layer with **MongoDB Vector Search** for embeddings and document storage, then containerized and deployed the service on **AWS (ECS/Fargate)**, maintaining **100+** concurrent requests at **p95 latency of 1–2 seconds**.